8.2 Proving Triangle Similarity by AA

Name _____ Period ____ Date _____

Geometry 8.2 (30-minute plan) · Student sheet

I can…

- use two pairs of congruent angles to show two triangles are similar.
- write a similarity statement with the vertices in the correct order.
- use similar triangles to find a missing length, including a height I can't measure directly.

Vocabulary · fill in the missing word

AA Similarity Theorem — if two angles of one triangle are _____ to two angles of another triangle, the triangles are similar

Indirect measurement — finding a length with similar _____ instead of measuring it

Reflexive Property — an angle is _____ to itself (a shared angle)

Vertical angles — opposite angles formed by two _____ lines; they are congruent

Part 1 · Notes: fill in the blanks

AA Similarity Theorem: if two _____ of one triangle are congruent to two _____ of another triangle, then the triangles are _____.

Congruent angles can come from a shared angle, _____ angles, or _____ lines cut by a transversal.

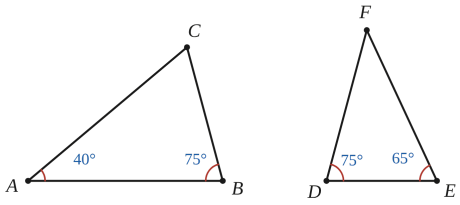
Write the similarity statement so the _____ angles are in the same positions.

Indirect measurement: use _____ triangles to find a length you cannot measure directly (like a height from shadows).

Part 2 · Examples (work them with the class)

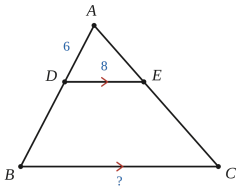
Example 1

Determine whether the triangles are similar. If they are, write a similarity statement.



Example 2

In △ABC, $\overline{DE} \parallel \overline{BC}$. Show that △ADE ~ △ABC. Then find BC when AD = 6, AB = 15, and DE = 8.



Example 3

A 6-foot-tall person casts a 4-foot shadow. At the same time, a flagpole casts a 30-foot shadow. How tall is the flagpole?

Part 3 · Partner practice

Partner practice

$\triangle PQR$ has $m\angle P = 90°$ and $m\angle Q = 35°$. $\triangle STU$ has $m\angle S = 55°$ and $m\angle T = 90°$. Are the triangles similar? If so, write a similarity statement.